

PRD — Paragon (CFA · Car Fluid Analyzer)

Version 1.0 · **Team** Chosun University — Qualcomm Institute Team A · **Sources** RESULTS.md, DATA_SUMMARY.md, EXPERIMENT_REPORT.md

1. One-line definition

A web-based AI tool that predicts a vehicle's **drag coefficient (Cd) in seconds** from its 3D shape or design parameters, and flags whether it clears a target. Not a replacement for CFD — a **fast pre-filter for early-stage design**.

2. Problem

- Aerodynamic drag drives fuel and electric-range efficiency; reducing drag alone yields large energy savings.
- CFD, the standard way to measure drag, is **slow and expensive**: days to weeks per high-fidelity design, supercomputer-scale compute, and a specialist team. (Building the DrivAerNet++ dataset alone took 2,880 CPU cores / 39 TB.)
- As a result, **rapidly comparing many candidate designs early is impossible**, and anyone who is not a CFD expert is locked out entirely.

3. Target users

Priority	User	Need
P0	Automotive designer / engineer	See a design's aerodynamic performance instantly in early stages, without CFD expertise
P1	Product manager	Compare a set of candidate designs to support decisions
P2	CFD engineer	Narrow the field before committing to full CFD

4. Core feature requirements

ID	Feature	Description	Priority
F1	3D file upload	Upload <code>.stl</code> / point cloud → sampling → preprocessing	P0
F2	Cd prediction	Return a Cd value within seconds of upload (PointNet surrogate)	P0
F3	Target check	Show over/under status against a user-set target Cd	P0
F4	Candidate ranking	Upload several designs and sort by lowest drag — what the model does best (rank accuracy 94.6 %)	P1
F5	Parameter prediction	Without a 3D file, enter design parameters → gradient-boosted / RF surrogate → instant Cd (lightweight track)	P1
F6	OOD warning	Detect out-of-distribution input (SUV/truck, etc.) and show a confidence flag	P1
F7	Design guidance	Suggest improvement directions when the target is exceeded (e.g. rear-end shape)	P2 (stretch)

5. Non-functional requirements

- **Response time:** end-to-end under 10 s from upload to result (including preprocessing; inference itself under 1 s).
- **Model lightness:** PointNet at 0.81 M parameters — small enough to serve on a CPU, not just a GPU.
- **Platform:** web interface (no install, browser access).
- **Input robustness:** defensive handling of unit-ambiguous STL (mm/m mix) and non-watertight meshes — the model depends on absolute metre scale, so unit estimation/validation is required.

6. Success metrics

The headline product metric is **pairwise ranking accuracy** — "given two designs, does it identify the lower-drag one?" is what the product actually does (50 % = a coin flip).

Stage	Goal	Status (official test, 1,158 designs)
MVP	Training pipeline complete	✔ Done
Tier 1	$R^2 \geq 0.50$	✔ 0.814 from dimension regression alone
Tier 2	$R^2 \geq 0.60$	✔ PointNet 0.968
Error	$MAE \leq 0.01$	✔ 0.0051 (5.1 drag counts)
Final	3D upload → PointNet → Cd web demo	✔ Shipped (point-cloud upload live)

Internal quality bar (global averages hide failures, so **per-body-type decomposition is mandatory**):

- Per-body-type $MAE \leq 7$ drag counts (current: Fastback 4.7 / Estate 5.5 / Notchback 6.6 ✔)
- Per-body-type ranking accuracy ≥ 85 % (current: 95.4 / 87.3 / 86.2 % ✔)
- Rationale: dimension-based models hit 51.9 % ranking accuracy on Estate (a coin flip) — the point-cloud model lifting that to 87.3 % is the core differentiator.

7. Out of scope

- **Absolute Cd certification/measurement** — Paragon is a relative-comparison pre-filter. Final validation belongs to high-fidelity CFD / a wind tunnel.
- Flow-field (velocity/pressure) visualisation — only the single scalar Cd is predicted.
- Reliable prediction outside the sedan family (SUV / truck / commercial) — training data is DrivAer sedan variants only.
- Phone LiDAR scan input — a future extension candidate (needs research on partial-scan and noise degradation).

8. Technical approach

Track 1 – ML

23 design parameters (CSV) → Gradient-boosted / RF → Cd

Track 2 – DL

STL / point cloud → FPS-2048 sampling → PointNet (0.81 M) → Cd

(evaluated against mesh-native RegDGCNN → PointNet adopted)

- **Data:** DrivAerNet++ (Elrefaie et al., NeurIPS 2024) — 7,713 sedan variants, 100k points each, CFD-computed Cd labels. Official split 5,398 / 1,157 / 1,158 (no re-splitting).

- **Design principles:** ① keep metre scale (absolute height is the strongest signal, $r = +0.83$); ② decompose evaluation by body type; ③ ranking accuracy as the headline metric.

9. Risks and mitigations

Risk	Impact	Mitigation
STL unit ambiguity (mm/m)	Scale-dependent model outputs a completely wrong value	Estimate units from bbox size on upload + user-confirmation UI
OOD input (non-sedan)	Confidently wrong answer	F6 OOD flag, show confidence interval
No real-vehicle Cd validation	Only simulation-internal performance is proven	Scan 5–10 real cars with manufacturer-published Cd as a validation set (never for training)
User mesh quality (holes, noise)	Preprocessing failure or degraded accuracy	Point-sampling pipeline needs no watertightness — reinforce with degradation-condition training (R3)

10. Open questions (team decisions)

1. Whether to include F7 (design guidance) in demo scope — saliency- or sensitivity-based; implementation difficulty is high.
2. Deployment form of the web demo (local GPU server vs cloud). → **Decided: single-service Docker on Render.**
3. Whether to expose Track 1 (parameter ML) as a product feature or keep it as an internal baseline. → **Decided: exposed as the parameter-slider studio path.**